

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA0301

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

***CO5:** Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Analytical Problem-Solving Assessment

Duration: 1hr

1. A smart city traffic management system uses **Multi-Agent Reinforcement Learning (MARL)** to coordinate traffic signals. Evaluate the **effectiveness of MARL** in improving traffic flow while considering scalability, coordination, fairness, and ethical challenges. .
(5 Marks)

1. Introduction

Traffic congestion is one of the major problems faced by modern cities. As the number of vehicles increases, fixed-time traffic signals often become inefficient because they cannot respond effectively to changing traffic conditions.

For example, one road may have heavy traffic while another road has very few vehicles. A fixed traffic signal may still provide equal green-light time to both roads, resulting in unnecessary waiting.

Multi-Agent Reinforcement Learning (MARL) provides an intelligent approach to this problem.

In MARL, each traffic intersection can be represented by an independent **RL agent**. Each agent observes the traffic conditions at its intersection, selects signal timings, receives rewards, and learns to improve traffic flow.

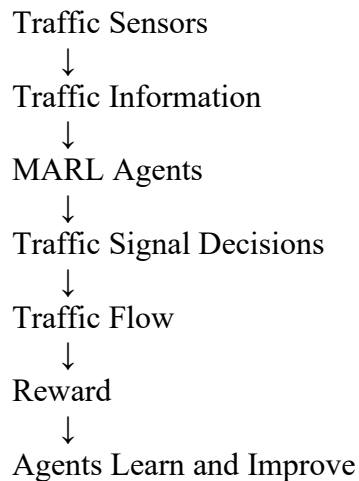
Instead of controlling every traffic signal independently, MARL allows the agents to **communicate and coordinate** with neighboring intersections.

Main Objective

The main objectives of MARL-based traffic management are:

- Reduce traffic congestion.
- Reduce vehicle waiting time.
- Reduce queue length.
- Increase traffic throughput.
- Coordinate neighboring traffic signals.
- Give appropriate priority to emergency vehicles.
- Improve overall road-network efficiency.
- Adapt to changing traffic conditions.

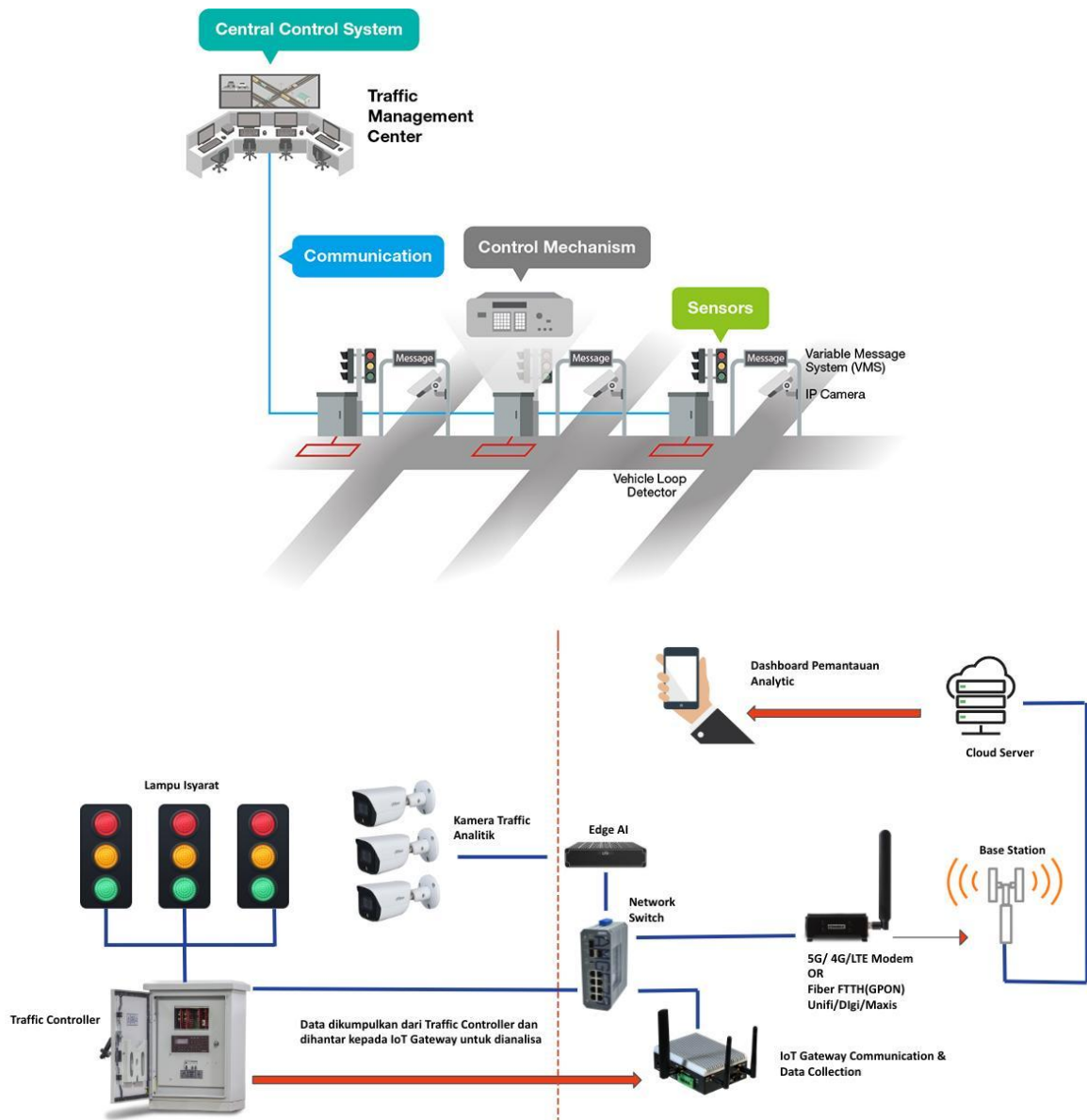
Basic Concept



SMART CITY TRAFFIC MANAGEMENT SYSTEM

2. System Architecture

A smart-city traffic management system can use sensors, cameras, communication networks, and MARL agents to control traffic signals.



Main Components

1. Traffic Sensors

Sensors detect:

- Number of vehicles
- Vehicle speed
- Queue length
- Lane occupancy
- Traffic density

2. Cameras

Cameras can provide visual information about traffic conditions.

3. Traffic Signal Controller

Controls:

- Red signal
- Yellow signal
- Green signal
- Green-light duration

4. MARL Agents

Each intersection has an RL agent responsible for making traffic-signal decisions.

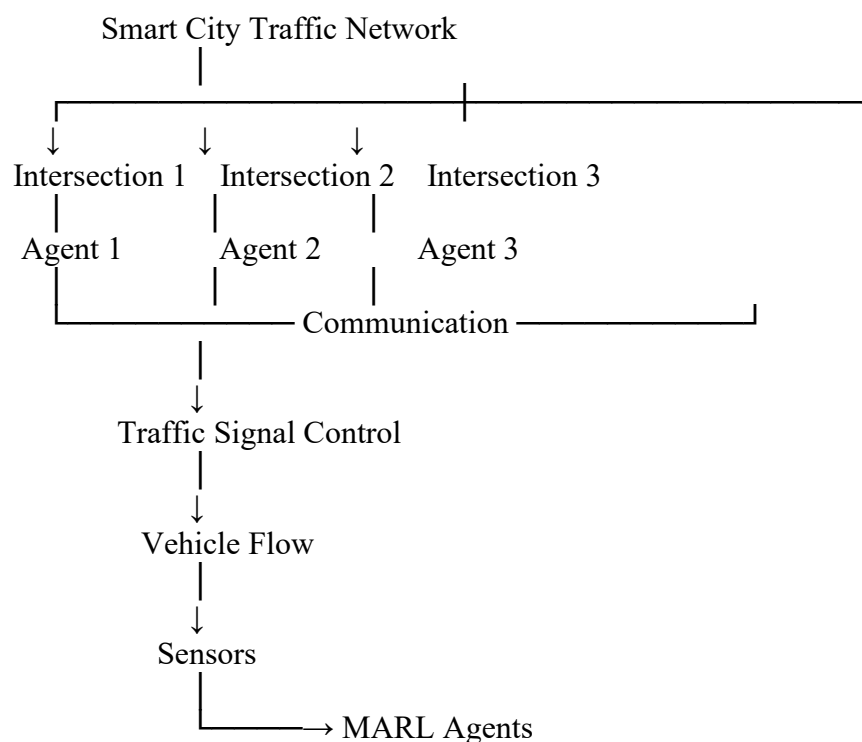
5. Communication Network

Agents can share information with neighboring intersections.

6. Central Traffic Management System

A central system can monitor the overall network and provide additional information or constraints.

Architecture



MARL CONCEPT

3. What is Multi-Agent Reinforcement Learning?

In normal Reinforcement Learning, one agent interacts with an environment.

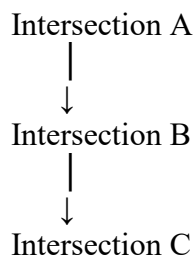
In **Multi-Agent Reinforcement Learning**, multiple agents interact with the same environment and may cooperate or compete with each other.

For traffic management:

- **Agent:** Traffic signal controller
- **Environment:** Road intersection
- **State:** Traffic condition
- **Action:** Signal-control decision
- **Reward:** Traffic-flow performance

Example

Consider three connected intersections:



Agent A controls Intersection A.

Agent B controls Intersection B.

Agent C controls Intersection C.

If Agent A gives a green signal without considering Agent B, vehicles may reach B and face a red signal, creating a queue.

Therefore, the agents should coordinate their decisions.

STATE, ACTION AND REWARD

4. State Space

Each traffic agent observes the state of its intersection.

A state can be represented as:

$$S_i = [q_i, v_i, d_i, p_i, t]$$

where:

- q_i = queue length
- v_i = vehicle speed
- d_i = traffic density

- p_i = current signal phase
- t = time

The agent may also receive information from neighboring intersections.

5. Action Space

The agent controls traffic signal decisions.

Possible actions include:

- Keep current phase.
- Change from red to green.
- Change from green to red.
- Extend green duration.
- Reduce green duration.

For example:

Action 1 → Keep Green

Action 2 → Switch to Red

Action 3 → Extend Green

Action 4 → Change Phase

6. Reward Function

The reward should encourage smooth traffic flow.

A simple reward function can be:

$$R_t = -(\alpha Q + \beta W + \gamma D)$$

where:

- Q = queue length
- W = waiting time
- D = traffic delay
- α, β, γ = weighting factors

A high reward means:

- Short queues
- Low waiting time
- Low congestion
- High traffic throughput

The system can also include penalties for:

- Excessive signal switching

- Blocking intersections
- Ignoring emergency vehicles
- Unfair waiting times

COORDINATION BETWEEN AGENTS

7. Why Coordination is Important

Independent traffic signals can make locally good decisions that produce poor results for the overall city.

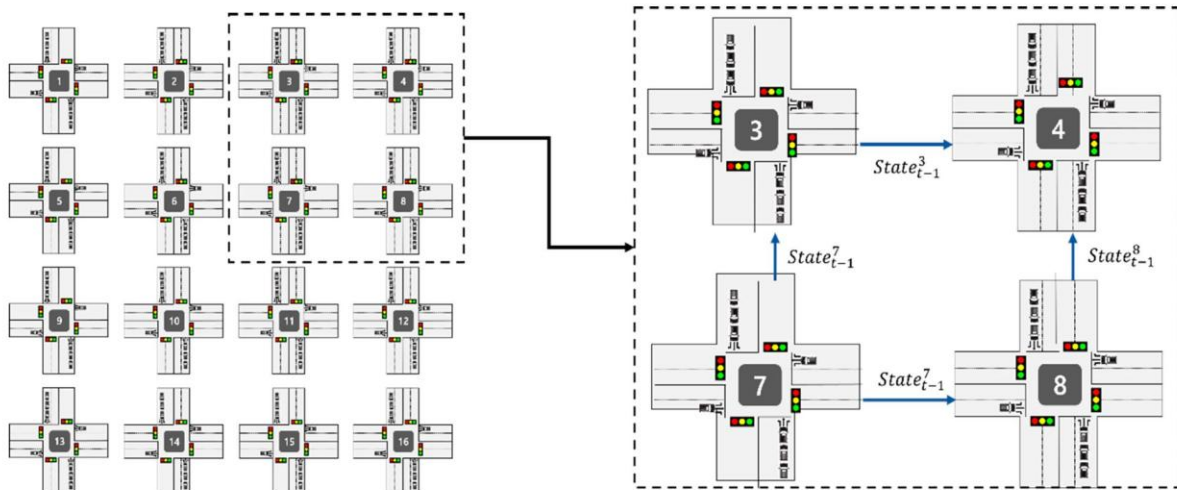
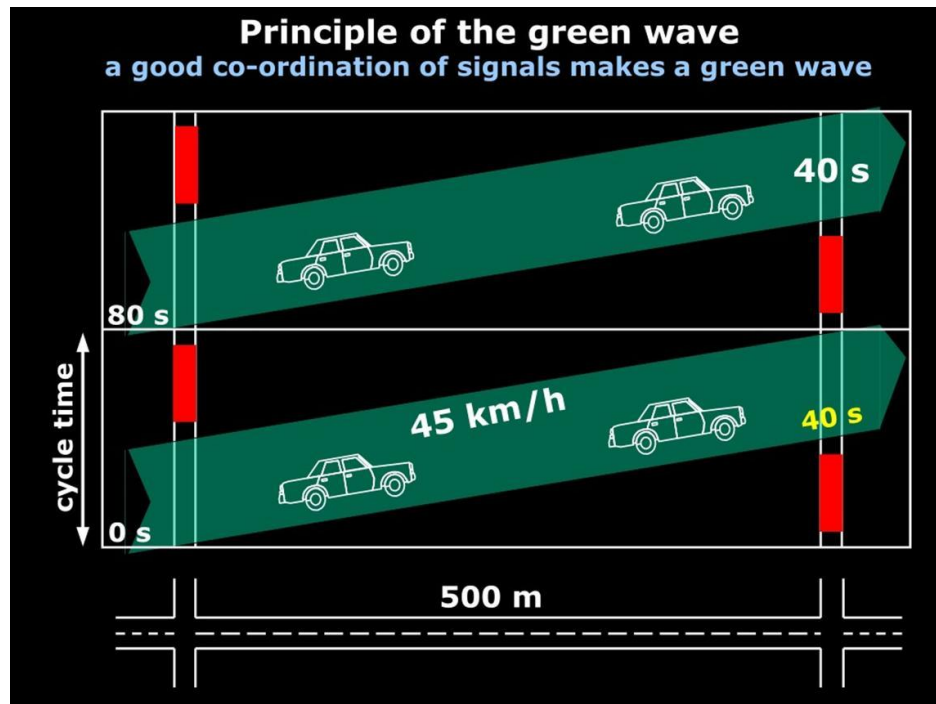
For example:

Intersection A → Green
↓
Large group of vehicles moves
↓
Intersection B → Red
↓
Large queue forms
↓
Congestion

MARL can coordinate the signals:

Intersection A → Green
↓
Vehicles move
↓
Intersection B → Green
↓
Vehicles continue
↓
Intersection C → Green
↓
Smooth Traffic Flow

This creates a **green wave**, where vehicles can pass through multiple intersections with fewer stops.



8. Types of Coordination

Centralized Coordination

A central controller receives information from all intersections and makes decisions.

Advantage: Global coordination.

Disadvantage: Can become computationally expensive as the city grows.

Decentralized Coordination

Each agent makes its own decision using local information.

Advantage: Highly scalable.

Disadvantage: Global coordination may be weaker.

Hybrid Coordination

Agents make local decisions while sharing selected information with neighboring agents.

This can provide a balance between scalability and coordination.

SCALABILITY

9. Scalability Challenge

A smart city may contain hundreds or thousands of traffic intersections.

If every intersection is controlled by an agent, the system must handle a very large number of agents.

For example:

Small Network

3 Intersections

↓

3 Agents

Medium Network

50 Intersections

↓

50 Agents

Large Smart City

1000 Intersections

↓

1000 Agents

As the number of agents increases, communication and computation become more difficult.

10. How MARL Handles Scalability

Parameter Sharing

Multiple agents can share the same neural-network parameters.

This reduces the amount of training required.

Hierarchical Control

Traffic can be controlled at different levels:

City Level

↓

Zone Level



Intersection Level

Local Observation

Agents can focus primarily on nearby intersections rather than the entire city.

Selective Communication

Agents exchange only important traffic information.

Clustering

Similar intersections can be grouped into zones.

11. Scalability Assessment

Approach	Scalability
Independent RL agents	High
Fully centralized MARL	Low–Medium
Decentralized MARL	High
Hierarchical MARL	Very High
Hybrid MARL	High

Therefore, **decentralized or hierarchical MARL** is generally more suitable for large smart-city networks.

FAIRNESS

12. Fairness in Traffic Management

Optimizing overall traffic flow does not automatically guarantee fairness.

Suppose a MARL system tries to minimize total waiting time. It might continuously prioritize a major road because it carries more vehicles.

As a result, vehicles on a smaller road may wait for a very long time.

Major Road



More Green Time

Minor Road



Long Waiting Time

This creates an unfair traffic-control policy.

13. Fairness-Aware Reward

The reward function can include fairness.

For example:

$$R = -(TotalDelay + \lambda FairnessPenalty)$$

A fairness penalty can increase when one road experiences significantly greater waiting time than others.

The system can therefore balance:

- Overall traffic efficiency
- Individual-road waiting time
- Pedestrian waiting time
- Public transport priority

4. Important Fairness Considerations

MARL should avoid:

- Continuously prioritizing major roads.
- Ignoring low-traffic areas.
- Excessive waiting on side roads.
- Discriminating against particular road users.
- Giving inappropriate priority based on biased data.

A good traffic system should optimize **overall efficiency while maintaining reasonable service for all road users**.

ETHICAL CHALLENGES

15. Ethical Issues

AI-based traffic management affects millions of people. Therefore, ethical issues must be considered.

1. Privacy

Traffic cameras and sensors may collect information about vehicles and people.

The system should avoid unnecessary collection of personally identifiable information.

2. Surveillance

Continuous camera monitoring may create concerns about excessive surveillance.

3. Bias

If training data represents some areas better than others, the system may develop biased decisions.

4. Emergency Vehicle Priority

Ambulances, fire engines, and police vehicles may require priority.

The system should ensure that emergency vehicles can pass safely.

5. Transparency

Traffic authorities should be able to understand why the system changed a signal.

6. Safety

An RL agent should never be allowed to make unsafe signal changes merely to maximize reward.

7. Accountability

There must be a responsible authority when an AI-controlled traffic system makes a harmful decision.

EFFECTIVENESS AND PERFORMANCE EVALUATION

16. Performance Metrics

MARL effectiveness can be measured using:

Average Waiting Time

$$AWT = \frac{\sum \text{Waiting Time}}{\text{Number of Vehicles}}$$

Lower is better.

Average Queue Length

Measures the average number of vehicles waiting at an intersection.

Throughput

Number of vehicles successfully passing through the network.

Higher throughput is better.

Average Travel Time

Measures how long vehicles take to reach their destinations.

Number of Stops

Lower number of stops generally indicates smoother traffic flow.

Fuel/Energy Consumption

Reduced idling and stopping can reduce fuel consumption.

17. Example Simulation Results

The following values are **illustrative results**:

Metric	Fixed-Time Signals	MARL
Average Waiting Time	95 sec	48 sec
Average Queue Length	28 vehicles	14 vehicles
Average Travel Time	18 min	12 min
Throughput	1,000 vehicles/hr	1,350 vehicles/hr
Number of Stops	7	4
Traffic Delay	High	Low

These results indicate that MARL can potentially improve traffic flow by adapting signal timing to real-time traffic conditions.

18. Traffic Flow Improvement

Average Waiting Time



MARL can reduce waiting because signal timings are dynamically adjusted based on traffic conditions.

ADVANTAGES, LIMITATIONS AND CONCLUSION

19. Advantages of MARL

Adaptive

MARL can respond to changing traffic conditions.

Distributed Control

Each intersection can make local decisions.

Coordination

Agents can cooperate with neighboring intersections.

Reduced Congestion

Dynamic signal control can reduce queue lengths and delays.

Scalability

Decentralized approaches can be expanded to larger networks.

Emergency Management

Priority can be incorporated for emergency vehicles.

Energy Efficiency

Reducing unnecessary stops and idling can reduce fuel consumption.

20. Limitations

Training Complexity

Training multiple interacting agents can be difficult.

Communication Overhead

Agents may need to exchange information.

Non-Stationarity

Each agent changes its policy while other agents are also learning.

Safety

Exploration during learning can produce undesirable actions if not constrained.

Data Requirements

Large amounts of traffic data or simulation experience may be required.

Generalization

A model trained for one traffic network may not automatically perform well in another city.

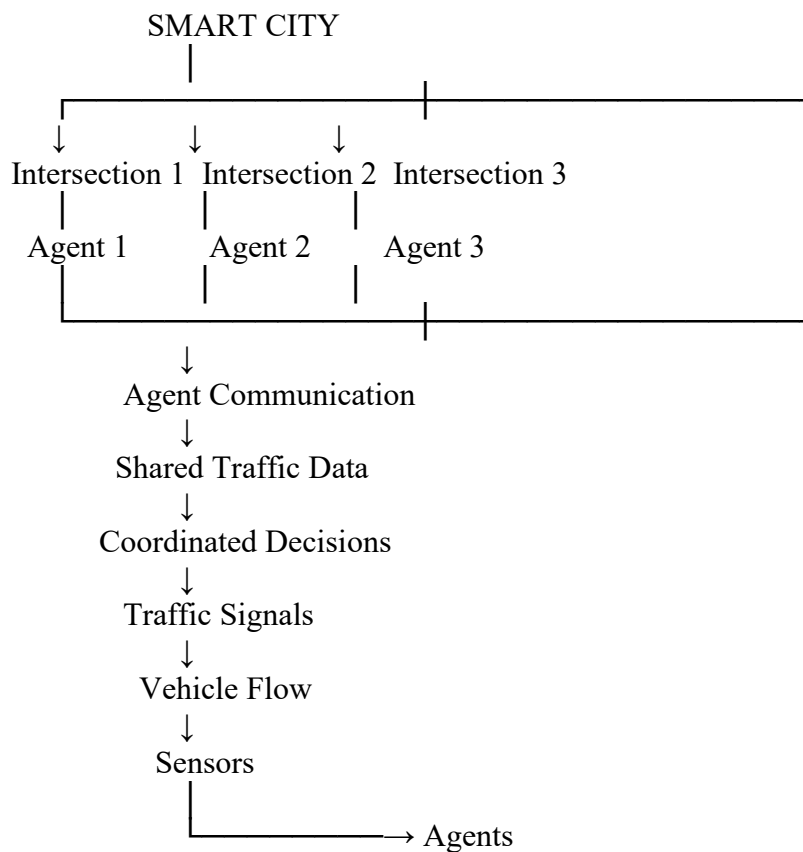
21. Overall Evaluation

Factor	MARL Effectiveness
Traffic-flow improvement	★★★★★
Adaptability	★★★★★
Coordination	★★★★★
Scalability	★★★★
Fairness	★★★
Safety	★★★
Privacy	★★★
Implementation complexity	★★

The ratings are qualitative and depend on the specific MARL design.

MARL is highly effective for **adaptive traffic-signal coordination**, but it must be carefully designed to address scalability, fairness, safety, privacy, and accountability.

22. Recommended MARL Architecture



23. Final Assessment

MARL is generally more adaptable than traditional fixed-time traffic-signal control because multiple agents can learn to respond to changing traffic conditions and coordinate their decisions.

However, the best MARL system should not optimize only traffic throughput.

It should simultaneously consider:

$$\boxed{\textit{Efficiency} + \textit{Coordination} + \textit{Scalability} + \textit{Fairness} + \textit{Safety} + \textit{Ethics}}$$

A practical solution is a **decentralized or hierarchical MARL system with constrained actions, fairness-aware rewards, privacy-preserving data collection, and human oversight.**

24. Conclusion

Multi-Agent Reinforcement Learning provides a promising approach for intelligent traffic signal management in smart cities. Each traffic intersection can act as an RL agent and learn how to control signal phases based on traffic conditions.

The major advantage of MARL is its ability to coordinate multiple intersections and adapt to changing traffic patterns. It can reduce waiting time, queue length, travel time, and congestion while increasing traffic throughput.

However, large-scale deployment introduces challenges. Scalability requires efficient decentralized or hierarchical architectures. Coordination requires communication between neighboring agents. Fairness requires preventing major roads from receiving excessive priority. Ethical challenges include privacy, surveillance, bias, safety, transparency, and accountability.

Therefore, MARL can be considered an effective technology for smart-city traffic management, provided that efficiency is balanced with fairness, safety, scalability, and ethical responsibility.

Code of Conduct:

I Malli Chandana(192472250) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Malli Chandana

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively